

Resolved-flux factor 2: limited response

Matched sensitivity | 21 September 2026 | GABLS1, 12.5 m, WENO9, one interior face, 300 s filter, 9 h, one seed.

Result: crediting twice the resolved transport reduces the added diffusivity, but does not restore resolved turbulence. In the final hour, first-level w^2 rises 7.1%, while peak w^2 falls 6.3% and vertically integrated resolved TKE falls 6.4%. Both factors remain far below the earlier no-closure near-wall variance.

The intervention: factor a multiplies only the signed local filtered resolved flux inside the closure deficit $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$. The same factor is used for momentum and heat. Factor 2 assumes an additional numerical flux equal and aligned with resolved transport. It neither doubles viscosity nor rescales diagnostic fluxes. The experiment does not measure numerical transport or establish that assumption.

Mechanism: final-hour first-face viscosity falls from 1.321 to 1.118 m^2/s (15.4%); heat diffusivity falls from 1.216 to 0.774 m^2/s (36.3%). The first-face SGS share of u-momentum transport remains large: 88.7% versus 86.8%. Reduced closure coefficients therefore produce only a modest change in the resolved/SGS partition.

Moments: at 12.5 m, w^2 changes from 0.008171 to 0.008754 m^2/s^2 , versus 0.06353 without interior closure. The third moment changes from $-1.872\text{e-}4$ to $-1.186\text{e-}4$ m^3/s^3 , and skewness from -0.253 to -0.145. Skewness is the ratio of hourly averaged moments, not an average of instantaneous skewness.

Mean profiles and exchange: u^* falls from 0.2886 to 0.2831 m/s and sensible heat flux changes from -15.54 to -15.38 W/m^2 . Against the fixed 1 m median at native heights $0 < z \leq 200$ m, final-hour RMS errors change by +0.3% for u , -9.2% for θ , and -7.3% for w^2 . Smaller profile errors coexist with lower resolved energy; they do not demonstrate restored turbulence.

Hour sensitivity matters: in 7–8 h, first-level w^2 instead falls 5.1%, integrated TKE falls 10.4%, and the temperature-profile error increases 8.0%. Peak variance and integrated energy decrease in both windows. The near-wall variance increase and temperature improvement in the last hour are not consistent across the two hours. One seed and neighboring hours do not provide an uncertainty estimate.

Comparison integrity: two fresh cases passed strict admission with matching source, grid, seed, initialization and exact 19-profile/541-series schedules. The factor-1 run reproduces every exported profile and time-series value from corrected array 7293's one-face300 case bit for bit. Native physical flux partition, support and coefficients were checked. Fixed 1 m medians are LES references, not observations.

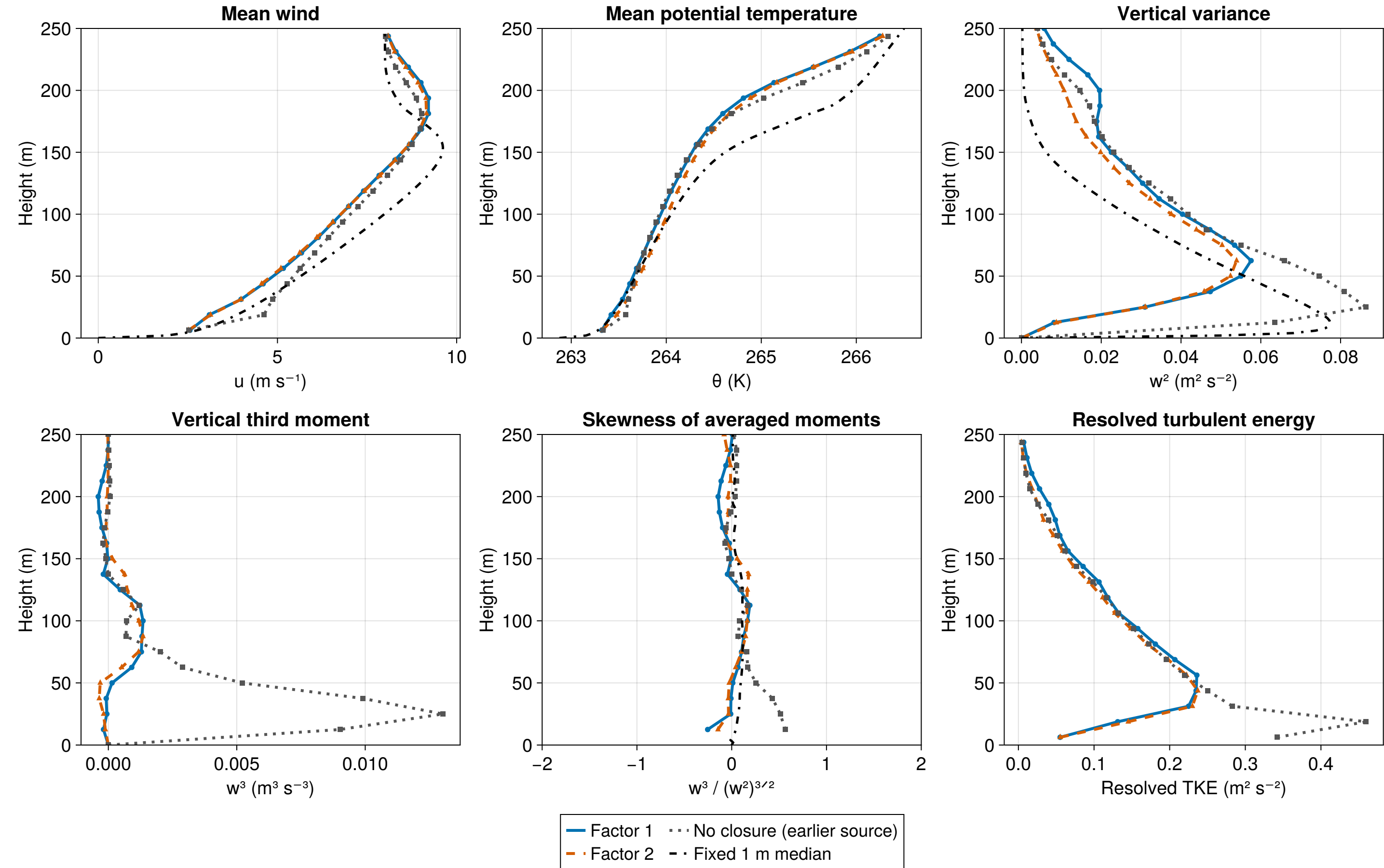
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Execution provenance: both model children completed 32400 s and exited zero. Slurm nevertheless marked the factor-2 batch exit as one. A separate harmless login-shell probe reproduces that pattern through failing `clear_console` in `.bash_logout` under `set-e`; actual batch shell depth was not recorded. The discrepancy and canceled downstream job are preserved. Admission relies on independently complete outputs, durable child exits, matching sources and logs, not a relabeled scheduler success.

Conclusion: this factor-2 test changes coefficients and exchange modestly, but provides no evidence of broad turbulence recovery. It does not identify the numerical flux or establish a calibrated correction factor. All earlier DYCOMS, GABLS1 and SurfaceLayerDiffusivity results remain in the master report. GABLS3's separate humidity-boundary validation is still pending; no new GABLS3 science is included.

Crediting twice the resolved flux: does turbulence recover?

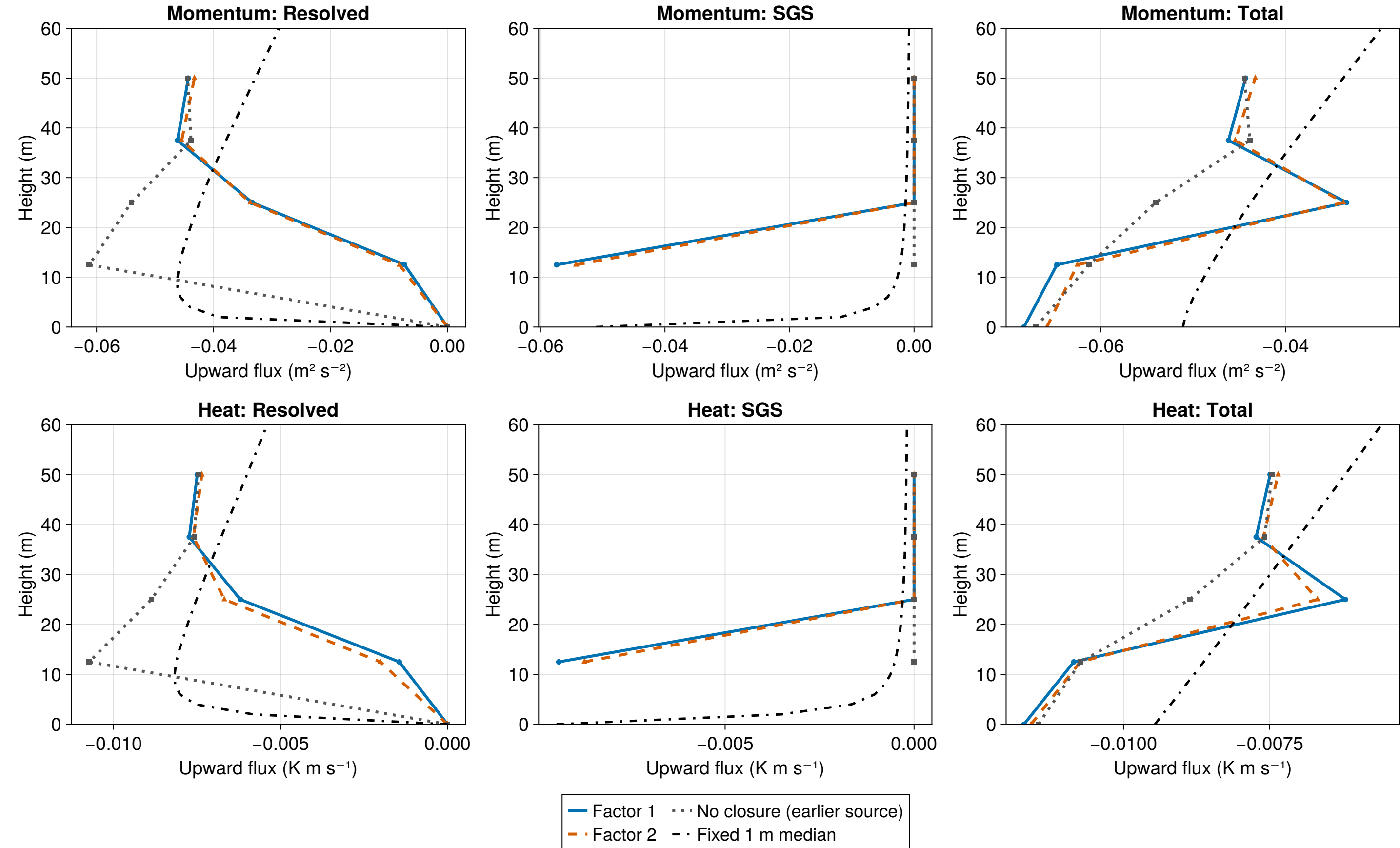
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor changes the closure deficit for momentum and heat; displayed physical fluxes are unscaled.
Earlier no-closure run is context. This tests an assumed numerical contribution, not a measured numerical flux.

Resolved and SGS transport near the surface

Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = resolved + SGS. At the wall, total includes the prescribed flux; artificial SGS wall zeros are omitted.
Each panel includes the fixed 1 m median wherever that reference quantity is available.